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1. Research Interests

- Vibrotactile perception and haptic interfaces
- Time-frequency analysis of skin vibrations
- Phase-based tactile information processing
- Multisensory UX design (haptics × audio × visual)
- Machine learning for human perception modeling
- Mechanism of physical sensing systems
- Usage and methodology of integrating multimodal interfaces into daily life

2. Education

Ph.D. Candidate, Scientific and Engineering Simulation

Nagoya Institute of Technology | 2023 – Present

Haptic Perception Laboratory | 2019 – Present

Supervisor | Prof. Yoshihiro Tanaka

- Research Topic:
“**Elucidation and establishment of methodologies of the spatiotemporal parameters affecting vibrotactile perception with body movement**”

Master of Engineering from the Creative Engineering Program

Nagoya Institute of Technology | 2021 – 2023

Bachelor of Engineering from the Creative Engineering Course

Nagoya Institute of Technology | 2017 – 2021

3. Research Experience

A. Skin Vibration Measurement / Phenomenon Discovery & Analysis

- Developed a complete pipeline for recording and analyzing skin vibrations during finger–surface interactions.
- Utilized high-speed acquisition (≥ 10 kHz) with NI USB-6218 and sensors (Showa Sokki 2302B, ADXL356/357).
- Discovered that tactile apparent motion can occur even when presented with the same stimulus through multiple body parts.
- Planned and conducted psychophysical experiments to evaluate the requirements for certain haptic phenomena.
- Performed calibration, denoising, and advanced time–frequency analysis (FFT/STFT/wavelet/Hilbert) including group delay and instantaneous frequency estimation.
- Investigated the perceptual contributions of magnitude versus phase spectra.

B. Phase-Only Texture Representation

- Constructed noise–texture chimera stimuli to demonstrate the strong perceptual effects of the phase spectrum.
- Conducted multivariate analysis (PCA/MDS/CCA) on large-scale phase-difference matrices.
- Analyzed temporal structures (30–250 ms windows) across participants.

C. Real-Time Haptic Signal Generation

- Developed multiple PyQt5-based GUIs for research and demonstration.
- Implemented real-time vibration synthesis (PyAudio) and integrated force/accelerometer sensor input with LED synchronization.
- Created real-time visualization tools (FFT, PSD, waveform).

D. Demonstration Experience for Various Audience

- Experience of demonstration for Top haptic conferences (WHC)
- Experience of demonstrating haptic illusions for general audience
- Experience of demonstration at the EXPO 2025@Osaka, Japan

E. Applied Haptics Projects

- Designed and demonstrated "Touch is Personal" Velcro texture tracing system.
- Advised the project to build a shared-perception system using inverse transfer functions and the efficiency of the system.
- Developed sports/entertainment vibration demos (fencing, dribbling experiences).

4. Technical Skills

Category	Skills
Programming	Python (NumPy, SciPy, scikit-learn, PyQt5, PyAudio), MATLAB, C++(Partially), Arduino
Signal Processing	FFT, STFT, wavelet, Hilbert transform, group delay, envelope extraction
Machine Learning / Statistics	PCA, MDS, CCA, Clustering, SVM, ANOVA, Nonparametric tests
Hardware & Experimental Systems (Prototyping/ Audio Interfaces)	NI USB-6218, Charge-type accelerometers, 6-axis force sensors, Smartphone vibrotactile actuators, AG06mk2, Play!3
Software Development	GUI (PyQt5), Real-time plotting, Real-time signal acquisition and rendering

5. Publications and Presentations

- T. Kuhara, K. Komazaki, J. Watanabe, and Y. Tanaka. (2025). Pseudo-Dribbling Experience Using Single Overlapped Vibrotactile Stimulation Simultaneously to the Hand and the Feet. *Multisensory Research*. 1-20. 10.1163/22134808-bja10157.
- T. Kuhara, H. Yukawa, and Y. Tanaka, (2024), "Exploring Vibrotactile Intensity Perception with Multiple Waveform Parameters," 2024 Asia Haptics, Sunway, Malaysia, 2024, doi:[10.48550/arXiv.2411.05099](https://doi.org/10.48550/arXiv.2411.05099)
- T. Kuhara, H. Yukawa and Y. Tanaka, (2025), "Influence of Longterm Duration and Damping Shapes to Perceived Intensity for Vibrotactile Stimulation," 2025 IEEE/SICE International Symposium on System Integration (SII), Munich, Germany, 2025, pp. 930-935, doi: 10.1109/SII59315.2025.10870919.
- T. Kuhara, A. Kawai, Y. Inoue, H. Yukawa, and Y. Tanaka, (2026), "Exploring Perceptual Effects of Phase Spectra in Vibrotactile Rendering", 2026 IEEE/SICE International Symposium on System Integration (SII), Cancun, Mexico, accepted as the proceedings
- *Additional manuscripts in preparation*: The perceptual and physical analysis of the effect of phase spectra in texture perception using vibrotactile stimuli

6. Awards and Achievements

- Selected as JST SPRING candidate for Nagoya Institute of Technology (2023)
- Demonstration presented at major exhibitions (e.g., *Expo 2025*, *WHC*, etc.).

7. Additional Experience

- Designed and executed tactile experiments with other universities and companies.
- Project: Tactile induced sports watching experience at All Japan Fencing Championship (in collaboration with NTT West).
- Developed sports-related tactile UX design proposals.